

PC5202 Advanced Statistical Mechanics

midterm test, closed book, 5 March 2011

1. Thermodynamics:

- Give at least two equivalent definitions of the chemical potential μ .
- Using Callen's maximum entropy postulates I, II, and III, show or argue that when two phases are in coexistence, e.g., ice in coexistence with liquid water, the chemical potentials of the two phases must be equal.
- State the Gibbs phase rule.

Ans:

$$a) \quad \mu = \left(\frac{\partial U}{\partial N} \right)_{S,V} = \left(\frac{\partial F}{\partial N} \right)_{T,V} = -T \left(\frac{\partial S}{\partial N} \right)_{U,V} = \frac{G}{N}$$

b) Let the two subsystems called 1 and 2, then we have

$$dS_1 = \frac{dU_1}{T_1} + \frac{P_1}{T_1} dV_1 - \frac{\mu_1}{T_1} dN_1,$$

$$dS_2 = \frac{dU_2}{T_2} + \frac{P_2}{T_2} dV_2 - \frac{\mu_2}{T_2} dN_2$$

Since the system as a whole are closed, we have $d(U_1+U_2)=0$, $d(V_1+V_2)=0$, $d(N_1+N_2)=0$, but total entropy S_1+S_2 must be a maximum with respect to the change of U_1 , V_1 or N_1 , i.e.,

$$d(S_1 + S_2) = \left(\frac{1}{T_1} - \frac{1}{T_2} \right) dU_1 + \left(\frac{P_1}{T_1} - \frac{P_2}{T_2} \right) dV_1 - \left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T_2} \right) dN_1 = 0,$$

we find $T_1=T_2$, $P_1=P_2$, and $\mu_1=\mu_2$.

- $f = r - M + 2$ where f is the number of degrees of freedom of intensive variables (varying of which do not lead to change of phases), r is the number of chemical components, and M is the number of coexisting phases. (see Callen's book, page 245-248).

2. The quantum-mechanical mixed state (density operator) is of the form

$$\rho = \sum_i w_i |\Psi_i\rangle \langle \Psi_i|.$$

- What condition do we must impose for ρ to be a valid density matrix?
- Give the equation that ρ satisfies.
- What is ρ in a grand canonical ensemble?

Ans:

- a) For the formula to represent a valid density matrix (read Chap 8 of K. Huang), one must have w_i a real number and $w_i \geq 0$, $\sum_i w_i = 1$.
 - b) The von Neumann equation, $\frac{d\rho}{dt} = (\hat{H}, \rho) = \frac{1}{i\hbar} [\hat{H}, \rho]$
 - c) $\rho = \frac{e^{-\beta(\hat{H} - \mu\hat{N})}}{\Xi}$. The grand partition function Ξ is fixed by $\text{Tr}[\rho] = 1$. The trace is over the Fock space.
3. Consider a collection of bosonic particles moving in a 1D parabolic trap such that the energy levels of a single particle are given by $E_n = (n+1/2)\hbar\omega$, $n=0, 1, 2, \dots$
- a. Compute the grand partition function Ξ as a function of temperature T and chemical potential μ .
 - b. What is a typical fluctuation in the number of particles if the trap contains one particle on average; what if there are 10^6 particles. Answer quantitatively based on result in part (a).

Ans:

a)

$$\begin{aligned}
 \Xi &= \text{Tr}[e^{-\beta(H_N - \mu N)}] \\
 &= \sum_{\{n_0, n_1, n_2, \dots\}} e^{-\beta \sum_{i=0,1,2,\dots} (\epsilon_i - \mu) n_i} \\
 &= \prod_{i=0}^{\infty} \sum_{n_i=0,1,2,\dots} e^{-\beta(\epsilon_i - \mu) n_i} \\
 &= \prod_{i=0}^{\infty} \frac{1}{1 - e^{-\beta(\epsilon_i - \mu)}}, \quad \epsilon_i = (i + \frac{1}{2})\hbar\omega
 \end{aligned}$$

b)

$$\begin{aligned}
 \langle N \rangle &= \frac{\partial \ln \Xi}{\partial(\beta\mu)} = \sum_{i=0}^{\infty} \frac{1}{e^{\beta(\epsilon_i - \mu)} - 1}, \\
 \langle N^2 \rangle - \langle N \rangle^2 &= \frac{\partial^2 \ln \Xi}{\partial(\beta\mu)^2} = \sum_{i=0}^{\infty} \frac{e^{\beta(\epsilon_i - \mu)}}{(e^{\beta(\epsilon_i - \mu)} - 1)^2} = \langle N \rangle + \sum_{i=0}^{\infty} \frac{1}{(e^{\beta(\epsilon_i - \mu)} - 1)^2}
 \end{aligned}$$

If the occupation number per state is low (high T), the second term can be ignored. We have that

$\langle N^2 \rangle - \langle N \rangle^2 \approx \langle N \rangle$. If we have 1 particle on average, the standard deviation (square root of the variance) in the particle number is of order 1 (percentage deviation is 100%). If we have 10^6 particles, the standard deviation (fluctuation) in the particle numbers is of order $\sqrt{N} \approx 10^3$ (0.1 percents).

4. Consider the problem of a mean-field theory of an infinite long one-dimensional Ising model, with the classical Hamiltonian (energy) given by $H = -J \sum_i \sigma_i \sigma_{i+1} - h \sum_i \sigma_i$, $\sigma = \pm 1$. Let us have the spins of site 1 and 2 treated explicitly and exactly and the rest of the spins of site $\dots, -2, -1, 0$, and $3, 4, 5, \dots$, replaced by average, $m = \langle \sigma_i \rangle$.
 - a. Derive the mean-field equations for the average of two spins $\langle \sigma_1 \rangle$ and $\langle \sigma_2 \rangle$.
 - b. If $\langle \sigma_1 \rangle = \langle \sigma_2 \rangle = m$, discuss if there is a phase transition at $h=0$, in the sense that $m > 0$ for $T < T_c$ and $m=0$ for $T > T_c$ with $T_c > 0$.

- a) The effective Hamiltonian is obtained by replacing all other spins by the average m : $H_{\text{eff}} = -J\sigma_1\sigma_2 - Jm\sigma_1 - Jm\sigma_2 - h\sigma_1 - h\sigma_2 + \text{const}$. The partition function (drop the constant term) is

$$Z = \sum_{\sigma_1, \sigma_2} e^{-\beta H_{\text{eff}}} = e^{\beta[2(Jm+h)+J]} + e^{\beta[-2(Jm+h)+J]} + 2e^{-\beta J}$$

$$m = \langle \sigma_1 \rangle = \langle \sigma_2 \rangle = \frac{1}{Z} \sum_{\sigma_1, \sigma_2} \sigma_1 e^{-\beta H_{\text{eff}}} = \frac{1}{Z} (e^{\beta[2(Jm+h)+J]} - e^{\beta[-2(Jm+h)+J]})$$

$$= \frac{\sinh[2\beta(mJ + h)]}{\cosh[2\beta(mJ + h)] + e^{-2\beta J}}$$

- b) Yes, there is a phase transition at nonzero T_c . Because if $\beta = 1/(k_B T)$ is large (low temperature T), the equation becomes approximately the same as the standard one discussed in class. One can draw two curves of the right and left side to see if there is an intersection point for m or not.

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